

Parametric studies on $\text{MmNi}_{4.7}\text{Fe}_{0.3}$ based reactor with embedded cooling tubes for hydrogen storage and cooling application

Alok Kumar^a, Nithin N. Raju^b, P. Muthukumar^{a,b,*}

^a Centre for Energy, Indian Institute of Technology Guwahati, Assam, India

^b Department of Mechanical Engineering, Indian Institute of Technology Guwahati, Assam, India

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ABSTRACT

The present study reports the parametric investigation conducted on 4 kg of $\text{MmNi}_{4.7}\text{Fe}_{0.3}$ under different absorption/desorption conditions. Considering the influence of heat transfer rate on reaction kinetics, a 55 embedded cooling tubes (ECT) reactor of enriched design was fabricated. In order to analyse the effect of H_2 supply pressure on the absorption rate, the pressure was varied from 10 to 70 bar, in steps of 10 bar, maintaining 25 °C absorption temperature and heat transfer fluid (HTF) flow rate at 24 lpm. At a supply pressure of 40 bar, the absorption temperature was varied from 5 °C to 25 °C to understand its effect on the absorption rate while the HTF flow rate was fixed at 15 lpm. Similarly, to understand the effect of temperature on dehydriding rate, the HTF was supplied at a constant flow rate of 15 lpm and its temperature was varied from 5 °C to 30 °C. At an absorption condition of 70 bar, 25 °C and 24 lpm, the bed effectively absorbed 45.984 g hydrogen within 676 s as compared to 3.929 g in 32 s at 10 bar supply. For a fixed absorption of 50 g, the dehydriding rate was higher for 30 °C desorption temperature with 47.729 g of hydrogen being desorbed in 388 s, producing a total cooling effect of 463 kJ. Such systems can be integrated with a stationary fuel cell with added advantage of cooling fuel cell stacks.

Introduction

Considering the wide range of hydrogen storage capabilities and thermal applications, metal hydride (MH) based thermal machines have gained deep insight and focus by various researchers, which is evident from the extensive development of these systems in the past two decades [1]. The thermal performance of such systems directly relies on hydride bed absorption/desorption kinetics, which directly depends on reaction bed collective heat and mass transfer physiognomies [2]. Several studies have contributed on various aspects of MH based thermal machines towards understanding these characteristics and in turn, improve their performance. The effects of shape distribution, particle size, and packing of bed on effective thermal conductivity was studied by Smith and Fisher [3] and they reported that with an increase in density, there is an increase in the thermal conductivity of the bed. Matsushita et al. [4] have carried out a similar study and reported that the effective thermal conductivity of the MH bed is inversely proportional to the porosity of the bed. The improvement in the thermal conductivity of the MH bed can be obtained by doping MH bed with high thermal conductivity

materials in appropriate proportion. Groll M. [5] analysed the effect of introducing different matrices like aluminium (Al) and magnesium (Mg) on thermal conductivity of MH bed. When beds with Mg-matrix, 5 wt.% Al and 17 wt.% Al were analysed, their respective effective thermal conductivity values were observed to be 10.8, 6.1 and 13.1 W/mK. Similarly, Guillemot et al. [6] found that copper foam is more significant as compared to aluminium mesh in enhancing the thermal conductivity of MH bed. However, introduction of dopant in the MH bed always leads to the depletion in the H_2 storage capacity of the bed. This is because of the significant amount of bed volume being occupied by the doping elements, which results in lower storage capacity of the MH bed. Similarly, Manai et al. [7] compared the hydrogen storage capacity of Ti-Mn using different bed configurations/designs through numerical study. The different bed configurations used in the study were loose alloy bed, compact alloy bed, compact alloy bed with steel fins and bed with compact alloy and stainless steel mixture. The study reported that, the bed configuration comprising of Ti-Mn compact and steel fins shown better absorption/desorption characteristics, as compared with other bed design. This is due to the improvement in the heat transfer characteristics within the alloy bed. Bhogilla et al. [8] reported novel reactor

* Corresponding Author: Dr. P. Muthukumar, Professor, Department of Mechanical Engineering, Indian Institute of Technology Guwahati, Guwahati - 781039, Assam, India., Tel: +91-361-2582673, Fax: +91-361-2690762

E-mail addresses: pmkumar@iitg.ac.in, pmkumariitg@gmail.com (P. Muthukumar).

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Nomenclature

C_p	Specific heat capacity (kJ/kg°C)
D	Inner diameter of ECT (m)
h	Effective heat transfer coefficient (W/m ² K)
HTF_a	Fluid flow rate during absorption (lpm)
HTF_d	Fluid flow rate during desorption (lpm)
L	Length of ECT (m)
m	Mass flow rate (kg/s)
Nu	Nusselt number
P_d	Absolute desorption pressure (bar)
P_s	Absolute supply pressure (bar)
Pr	Prandtl number
Q	Rate of heat transfer (kW)

Re	Reynolds number
T_a	Absorption temperature (°C)
T_d	Desorption temperature (°C)
ΔT	Change in temperature (°C)

Abbreviation

ECT	Embedded cooling tubes
HTF	Heat transfer fluid
MH	Metal hydride
PCM	Phase change material

Subscripts

t	At any given time 't'
HTF	Heat transfer fluid (water)

design, wherein coil heat exchanger arrangement is used within the cylindrical MH reactor filled with $Ti_{0.98}Zr_{0.02}V_{0.43}Fe_{0.09}Cr_{0.05}Mn_{1.5}$. During absorption, faster hydriding rate was observed near the coil heat exchanger, which gradually spread over other regions in the MH bed. Lewis et al. [9] studied heat and mass transfer characteristics during absorption/desorption in a $LaNi_5$ based MH reactor with phase change material (PCM) incorporated with metal foam. A significant improvement in the hydriding/dehydriding kinetics was obtained in case of PCM with metal foam, as compared to PCM without metal foam.

To promote rapid heat transfer, the bed thickness of MH should be small, as it is one of the significant influential factors for faster reaction kinetics [10]. Anbarasu et al. [11] have explored the reactor design with cooling tubes and predicted the performance of $LmNi_{4.91}Sn_{0.15}$ based hydrogen storage reactor under various operating conditions. They observed improved reaction kinetics in the MH bed with increase in the

supply pressure and/or effective thermal conductivity. Askari et al. [12] studied the MH bed dynamic with multiple reactor configuration of cylindrical tanks. They observed that implementing heat transfer augmentations in design such as fins and concentric tubes with cold fluid flowing resulted in 80% enhancement of the storage time over reactor bed with no such modifications. Recently, Raju et al. [13] proposed a design methodology for the MH reactor with embedded cooling tube (ECT) arrangements. They factored in improved reaction kinetics while balancing it against parasitic thermal mass and ease in manufacturability of the reactor. These studies conclude that the reactor bed design and enhancement of heat transfer within the bed lead to better reaction kinetics and this results in less hydrogen charging/discharging time.

Because of the different alloy compositions of MH, their operating ranges vary significantly as well. The selection of suitable alloy compositions should be according to the intended application [14]. Within

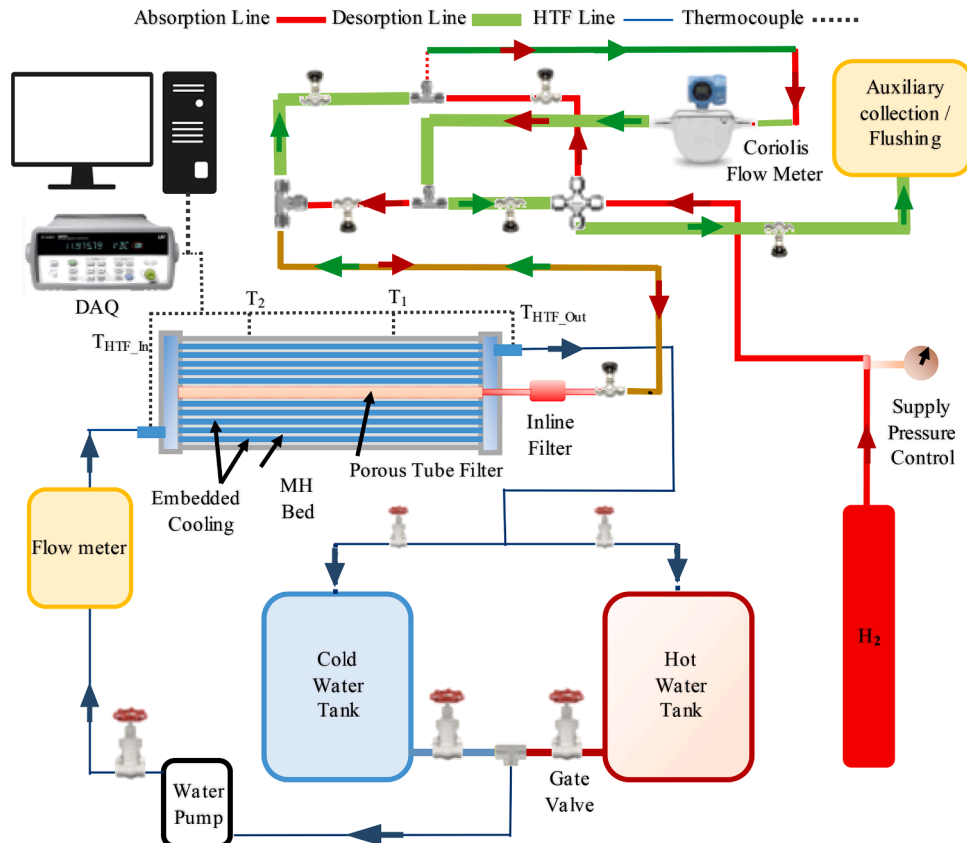


Fig. 1. Schematic view of experimental setup.

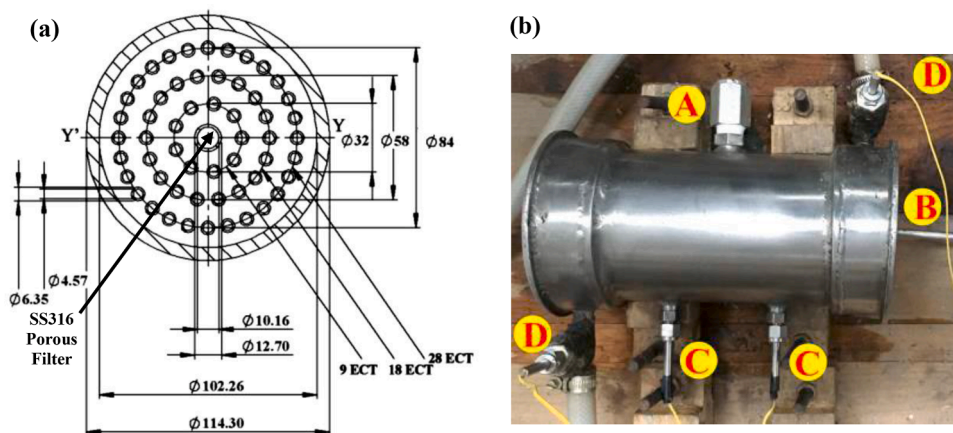


Fig. 2. Reactor configuration - (a) schematic view of cooling tubes arrangement and (b) pictorial view of the reactor (all dimensions are in mm).

the operating range of the selected MH alloy, suitable operating conditions of the bed can be obtained through a systematic parametric study by varying the experimental parameters like supply pressure of hydrogen, absorption temperature, desorption temperature, HTF flow rate within their operating ranges [15–19]. Raju et al. [15] reported the effects of HTF flow rate and supply pressure of H_2 on absorption, and HTF flow rate and bed temperature on the desorption kinetics of $LaNi_{4.7}Al_{0.3}$ based MH based H_2 storage system. According to the study, at higher supply pressure and lower absorption temperature, the rate of hydrogen absorption was faster because of the high driving force. Similar study was reported by Kumar et al. [16] on industrial scale reactor with 99 ECT having 40 kg alloy capacity. The study was carried out for 40 kg of $LaNi_{4.7}Al_{0.3}$, which was capable of storing nearly 550 g of hydrogen (~ 1.4 wt.%). Muthukumar et al. [17] studied MH based compressor driven cooling system for quasi-continuous cold generation. They obtained maximum coefficient of performance (COP) and specific cooling power (SCP) of 2.2 and 53.5 W/kg respectively, for 8 lpm cold HTF flow rate and 8 min cycle time. They further observed that the COP and SCP were directly proportional to the refrigeration temperature and inversely proportional to the sink temperature. Similarly, Marinelli et al. [20] reported different MH alloys for stationary applications, wherein they reported $LaNi_5$ and $TiFe_{0.85}Mn_{0.15}$ as most suitable alloy for stationary applications such as fuel cell, hydrogen storage, etc. Balcerzak [21] reported hydrogenation properties of nanostructured Ti-Zr-Ni. Using heat treatment and mechanical alloying (ball milling) under argon atmosphere at 750 °C for 30 min., nanocrystalline $Ti_{2-x}Zr_xNi$ ($x = 0.25, 0.5$) were produced. The result showed that, greater amount of Zr, i.e. for $Ti_{1.5}Zr_{0.5}Ni$ hydrogen storage capacity was maximum i.e. 2.17 wt.%. Zhu et al. [22] reported the synergistic effect of NbF_5 and porous boron nitride on the reversible dehydrogenation characteristics of MgH_2 . The study reports that there was 140 °C drop in the dehydrogenation temperature of MgH_2 , by using NbF_5 and porous boron nitride as dopant in the MgH_2 . Moreover, a significant reduction in the dehydrogenation time was also observed.

Though several studies have been reported on the various engineering applications of the MH based hydrogen storage system and their performance variation with respect to different operating conditions, most of these studies are restricted to computational/numerical approach and addressing the characterization of the MH alloys. Experimental investigations available in the literature are limited comparative to the extensive numerical studies. The necessity for establishing the operating conditions of suitable MH alloys filled in reactors with enhanced heat transfer augmentations is the motivation for the present work. The parametric study on $MmNi_{4.7}Fe_{0.3}$ filled in reactor with 55 ECT configuration has been carried out targeting the constant supply of hydrogen to a fuel cell at low temperature while simultaneously cooling the fuel cell stack. This system can be used for both stationary and

automobile applications, as its integration can ensure the continuous supply of hydrogen to the fuel cell stack for generating power with an advantage of additional cooling output.

Experimental Facilities and Procedure

Experimental facilities comprise of MH reactor, Coriolis mass flow meter, data acquisition system integrated with computer, HTF flow meter, hot water tank, cold water tank, centrifugal pump, hydrogen cylinder with flow regulator, inline filter, Swagelok SS316 tubes and valves arrangements for hydrogen transfer to/from reactor and suitable arrangements for HTF transport during absorption/desorption process as shown in Fig. 1. The shell and tube type MH reactor has a length of 160 mm and was fabricated using SS316. The reactor comprises of 55 embedded cooling tubes (ECT) which are arranged uniformly in three layers to provide enhanced heat transfer during the absorption/desorption process. The design of reactor and cooling tube arrangements are primarily based on the design methodology proposed by Raju et al. [13]. The dimensional specification of the reactor and the arrangement of ECT in each stack is presented in Fig. 2 (a). The shell side was filled with the MH alloy while the HTF flows through the tube side. The hydrogen transfer to/from the MH bed takes place through the cylindrical SS316 sintered porous tube of 2 μm porosity placed at the centre of the reactor as shown in Fig. 2 (a). Each ECT has a length of 160 mm with an outer diameter of 6.35 mm and a thickness of 0.89 mm. This effectively results in an overall heat transfer area of 0.126 m². The intermediate distance between the outer surfaces of ECT is found to vary between 3 mm and 4.8 mm in angular direction, while in radial direction, it varies between 5.9 mm and 6.6 mm.

As depicted in Fig. 2 (b), MH alloy is filled through the alloy filling port (A), the hydrogen is transferred to/from the reactor through the hydrogen supply tube (B), K-type thermocouples are attached to the reactor through weld fittings (C), and HTF inlet and outlet thermocouples are attached using fittings (D). The amount of H_2 absorbed/desorbed during absorption/desorption half cycle was measured using a Coriolis mass flow meter whose accuracy of measuring hydrogen flow rate is 0.0001 g/s and uncertainty in the measurement is $\pm 0.35\%$. The accuracy of K-type thermocouples used for measuring temperature is ± 0.5 °C and their data had been acquired by a data acquisition system at an interval of 1 s. Hydrogen supply pressure was set using SS316 high pressure gas regulator attached with hydrogen cylinder. The complete hydrogen flow circuit comprised of 1/4 inch Swagelok tubes with suitable valves as per schematic. Water was used as HTF during experiments wherein, cold and hot water respectively during absorption and desorption process was supplied from constant temperature baths at required flow rate using centrifugal pumps. The amount of HTF circulated during the experiments was recorded using a turbine type flow

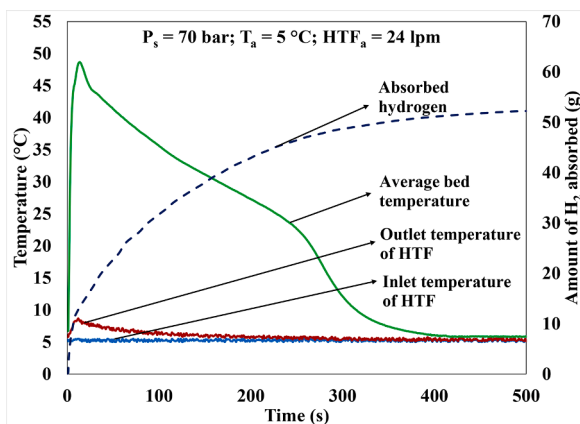


Fig. 3. Amount of H_2 absorbed and variation of bed temperature during final absorption cycle (P_s : Supply pressure; T_a : Absorption temperature; HTF_a : HTF flow rate during absorption)

meter. A secondary SS316 inline filter of 2 μm had been attached with reactor as an added deterrent to prevent the MH alloy powder from leaving the reactor. For a given half cycle, at any given time 't', the rate of heat transfer (kW) during absorption/desorption process is Q_t , which is obtained by:

$$Q_t = m_{HTF} * C_{pHTF} * \Delta T_{HTF} \quad (2.1)$$

Where, ΔT_{HTF} is the difference in temperature ($^{\circ}C$) between the HTF outlet and inlet at time 't' and C_{pHTF} represents the specific heat capacity of HTF (4.187 kJ/kg $^{\circ}C$). The mass flow rate of HTF m_{HTF} (kg/s) was obtained from flow rate of HTF (HTF_a/HTF_d) and the water density has been considered as 1000 kg/m 3 . The uncertainties in the experimental measurements were estimated using the method adopted by Kline and McClintock [23]. The uncertainty in measuring HTF flow rate and H_2 supply pressure was $\pm 1.67\%$ and $\pm 2.0\%$, respectively while the uncertainty in assessment of H_2 storage capacity was $\pm 2.5\%$.

Reactor activation

The fabricated 55 ECT reactor was filled with 4 kg of $MmNi_{4.7}Fe_{0.3}$. While preparing the system for activation, the reactor and the gas flow circuit have been made leak proof through systematic leakage testing by using argon at 100 bar supply pressure. The system was in this pressurised condition for nearly 48 h, to ensure leak-proof system. Then the system was evacuated down to 10^{-3} mbar vacuum pressure. The evacuated system was then flushed with H_2 gas at 5 bar which ensured the absence of any other gases in the bed. The bed was then evacuated while being maintained at $60^{\circ}C$. Further, H_2 was supplied at 50 bar while the HTF was at $30^{\circ}C$ and 24 lpm. In the first activation absorption cycle, 1.3213 g of hydrogen was absorbed. The bed was evacuated down to 10^{-3} mbar at $90^{\circ}C$ and the absorption process was repeated in a cycle. Even after 12 activation cycles, the bed absorbed just 10.8180 g of hydrogen. Hence, further activation was attempted at $5^{\circ}C$ by maintaining the hydrogen supply at 70 bar. Significant improvement in the absorption was observed and in 31st absorption cycle, the alloy reached a peak H_2 storage capacity of 1.36 wt.%.

The temperature distribution in the MH bed and HTF temperature (inlet and outlet) during this 31st activation cycle, and the quantity of H_2 absorbed are presented in Fig. 3. The MH bed reached the peak temperature of $48.6^{\circ}C$ within 20 s of the absorption cycle, and in the initial 250 s, it had absorbed about 48 g of hydrogen (~ 1.2 wt.%), which further reached to 54.413 g (1.36 wt.%) in 503 s.

Table 1
Parameters for Experimental Study

Half cycle	Investigation	Parameters	Values
Absorption	Effect of varying supply pressure	H_2 supply pressure (P_s)	10, 20, 30, 40, 50, 60 and 70 bar
		Temperature (T_a)	$25^{\circ}C$
		HTF flow rate (HTF_a)	24 lpm
		H_2 supply pressure (P_s)	40 bar
Desorption	Effect of varying temperature on desorption	Temperature (T_a)	5, 10, 15, 20 and $25^{\circ}C$
		HTF flow rate (HTF_d)	15 lpm
		Desorption pressure (P_d)	1 bar
		Temperature (T_d)	5, 10, 15, 20, 25 and $30^{\circ}C$
		HTF flow rate (HTF_d)	15 lpm

Experimental Parameters

In order to find out the suitable absorption/desorption operating conditions for the MH based system, a systematic parametric study has been carried out on the lab scale reactor filled with 4 kg of $MmNi_{4.7}Fe_{0.3}$. During experiments, hydrogen supply pressure and MH bed temperature were varied and the absorption/desorption performances of the bed were observed. The parameters considered in the present experimental study are summarized in Table 1. These parameters have been considered based on the operational feasibility and the targeted application.

The experiments were carried out at two different HTF flow rates of 24 lpm and 15 lpm. The average velocity of fluid through a single ECT has been estimated to be 0.44 m/s and 0.28 m/s, respectively for the flow rates of 24 lpm and 15 lpm. In the given parametric range, the fluid flow has been analysed to be a developing flow in nature. Nusselt number (Nu) has been estimated using the correlation proposed by Stephan and Preusser [24,25]. For given range of parameters, the effective heat transfer coefficient (h) was observed to vary in the range of 1920 to 2140 W/m 2K for flow rate of 24 lpm. Similarly, for flow rate of 15 lpm, h value was estimated to vary in the range of 1610 to 1820 W/m 2K .

Results and Discussions

Operating conditions play an important role in controlling the performance of MH based thermal systems. In order to assess the effect of different operating parameters on the charging/discharging characteristics of the MH bed, parametric study has been conducted on 55 ECT reactor filled with 4 kg $MmNi_{4.7}Fe_{0.3}$. For absorption study, the supply pressure was varied from 10 to 70 bar in steps of 10 bar, while the HTF temperature was maintained at $25^{\circ}C$ with a flow rate of 24 lpm. Further, to study the significance of absorption temperature, it was varied from $5^{\circ}C$ to $25^{\circ}C$ in steps of $5^{\circ}C$ with a constant HTF flow rate of 15 lpm while maintaining the hydrogen supply pressure constant at 40 bar. The amount of hydrogen absorbed and the bed temperature variation were analysed to compare the absorption performance. During the desorption study, at a HTF flow rate of 15 lpm, the desorption temperature was varied from $5^{\circ}C$ to $30^{\circ}C$ while the reactor was desorbed to a constant pressure collection chamber maintained at 1 bar. In order to compare the effect of desorption temperature, the amount of hydrogen absorbed in the reactor before desorption study was fixed at 50 g. Along with the amount of hydrogen desorbed and the bed temperature variation, the cooling produced during the desorption cycle was also analysed.

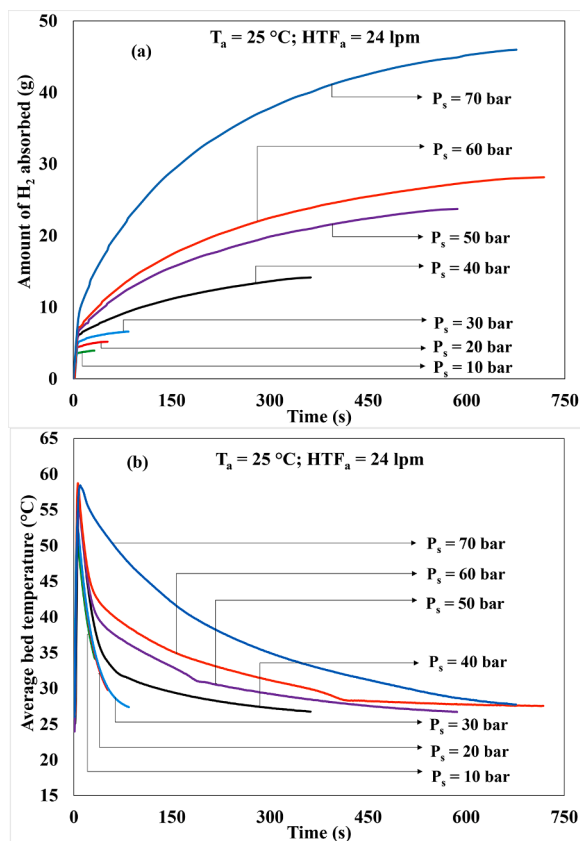


Fig. 4. Variation in (a) amount of H₂ absorbed and (b) average bed temperature by varying H₂ supply pressure during absorption. (P_s : Supply pressure; T_a : Absorption temperature; HTF_a: HTF flow rate during absorption)

Effect of varying H₂ supply pressure on absorption performance

As depicted in Fig. 4 (a) and 4 (b), the supply pressure (P_s) played a very prominent role in the amount of hydrogen absorbed and in the average bed temperature variation. For a supply pressure (P_s) of 10 bar, the amount of hydrogen absorbed was very less, i.e. 3.929 g in 32 s, which amounted to just 0.098 wt.%.

On increasing the supply pressure (P_s) further to 20 bar, 30 bar, 40 bar, 50 bar and 60 bar, the amount of hydrogen absorbed was reached to 5.1763 g (0.13 wt.%), 6.5771 g (0.16 wt.%), 14.1548 g (0.36 wt.%), 23.7296 g (0.59 wt.%) and 28.0845 g (0.70 wt.%), respectively in corresponding half cycle duration of 52 s, 84 s, 362 s, 586 s and 700 s. This signifies gradual improvement in the absorption capacity of the alloy with increasing supply pressure. For P_s = 70 bar, a very significant lift in the hydrogen storage capacity was observed. Within 676 s, 45.9844 g (1.15 wt.%) of hydrogen was absorbed and the bed attained a peak temperature of 57.8 °C in just 7 s.

The significant improvement in the absorption kinetics with increase in the H₂ supply pressure is due to the increase in the pressure gradient in between equilibrium bed pressure of MH alloy and the supply pressure, which indeed acts as the driving force for the boost up in the hydriding rate. Though the hydrogen absorption is significant at higher supply pressure, the parameter was not increased beyond 70 bar considering the operational feasibility and structural stability of the system.

Effect of varying temperature on absorption performance

Similar to supply pressure, variation in absorption temperature (T_a) also plays a decisive role on the absorption kinetics of the chosen alloy, as observed from Fig. 5 (a) and 5 (b). The experiments were conducted at

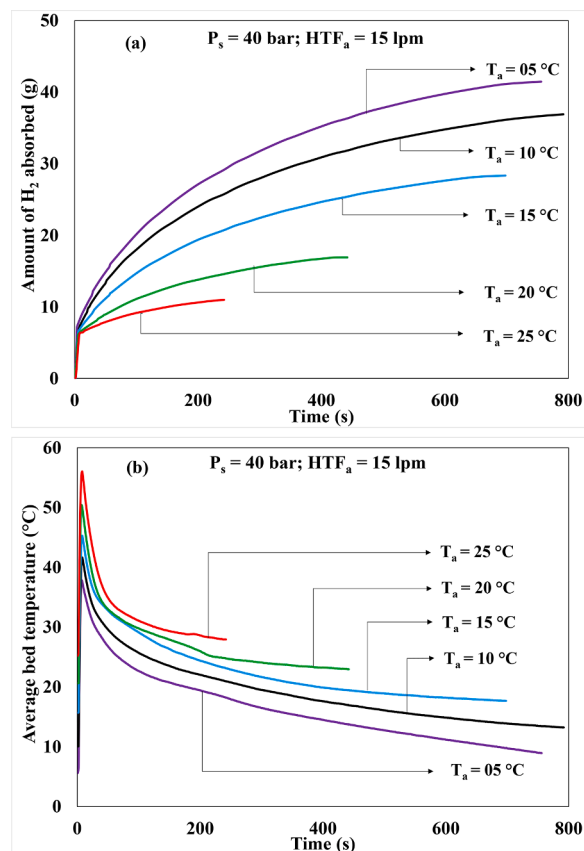


Fig. 5. Variation in (a) amount of H₂ absorbed and (b) average bed temperature by varying absorption temperature. (P_s : Supply pressure; T_a : Absorption temperature; HTF_a: HTF flow rate during absorption)

a fixed H₂ supply pressure (P_s) of 40 bar and varying the absorption temperature (T_a) from 5 °C to 25 °C in steps of 5 °C. For higher absorption temperature, the amount of hydrogen absorbed and the rate of absorption are very low. This is because of the lesser pressure gradient between the supply pressure (P_s) and the equilibrium bed pressure at higher temperature for the present MH alloy. But as the absorption temperature decreases, a substantial improvement in the absorption kinetics was observed. At 25 °C of absorption temperature (T_a), 10.979 g (0.27 wt.%) of hydrogen was absorbed in 242 s, with the bed attaining a peak temperature of 56.3 °C in just 8 s. Further, when the absorption temperature (T_a) was decreased to 20 °C, 15 °C, 10 °C and 5 °C, the respective amount of hydrogen absorbed increased to 16.905 g (0.42 wt.%), 28.342 g (0.71 wt.%), 36.89 g (0.71 wt.%) and 41.664 g (1.03 wt.%) in corresponding durations of 442 s, 698 s, 792 s and 756 s. This improvement in the absorption performance at lower temperature is because of the lower equilibrium pressure of bed. Thus, lowering of equilibrium bed pressure at lower absorption temperature resulted in higher pressure gradient between the supply pressure and equilibrium bed pressure, which finally led to rapid absorption kinetic of the MH bed. Though absorption is commendable at lower absorption temperature, maintaining it in the practical applications might prove to be energy intensive.

Effect of varying temperature on desorption performance

The desorption half cycles were performed for fixed amount of hydrogen absorbed in the bed i.e. 50 g, and the desorbed hydrogen was released to collection chamber maintained at 1 bar. During the desorption, supply of heat to the MH bed through the HTF results in elevated equilibrium pressure of the bed, and the pressure gradient

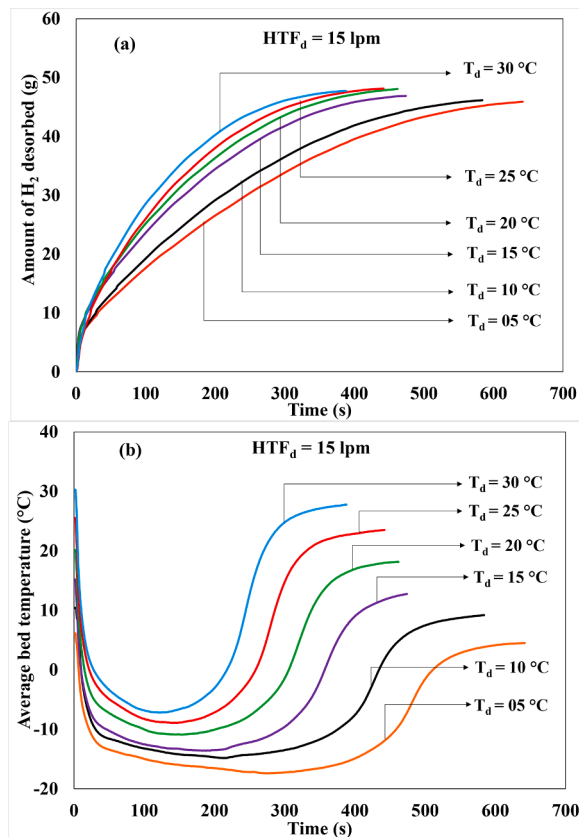


Fig. 6. Variation in (a) amount of hydrogen desorbed and (b) average bed temperature by varying desorption temperature. (T_d : Desorption temperature; HTF_d : HTF flow rate during desorption)

between this equilibrium bed pressure and the outlet pressure leads to faster desorption rate of hydrogen, when the valve to the collection chamber is opened. The noteworthy desorption characteristics of this alloy in 55 ECT reactor is the highlight of this study. As depicted in Fig. 6 (a) and 6 (b), for fixed absorption of 50 g, the reversibility of the alloy was appreciably near 96% for desorption temperature (T_d) in the range of 20 °C to 30 °C. There was slight decrease in the reversible desorption characteristics of the alloy at 5 °C, 10 °C and 15 °C, wherein the respective amount of desorbed hydrogen was 45.896 g, 46.162 g and 46.894 g. The rate of desorption decreased at lower temperature due to the reduced pressure gradient between bed and collection chamber.

The rapidity of the desorption reaction is evident from the half cycle time which was observed to be 642 s, 584 s, 474 s, 462 s, 442 s and 388 s respectively at the desorption temperature of 5 °C, 10 °C, 15 °C, 20 °C, 25 °C and 30 °C. The appreciable cooling effect that is produced during the desorption is due to the endothermic heat removed from the bed and HTF. The heat removal in MH bed can be observed from the prominent drop in average bed temperature depicted in Fig. 6 (b). The bed temperature dropped to -17.4 °C in 274 s when HTF was supplied at 5 °C and the HTF flow rate was 15 lpm. However, when the HTF was supplied at 30 °C there was a temperature drop of 37.2 °C within 123 s. This significant temperature drop in the bed is because to the faster desorption kinetics caused due to the significant pressure gradient between the equilibrium bed pressure and the desorption environment of 1 bar.

5.4. Effect of desorption temperature on cooling effect produced

The amount of cooling produced and the rate of cooling at different desorption temperature have been presented in the Fig. 7 (a) and 7 (b). It can be observed that the cumulative cooling effect produced in the temperature range of 15 °C to 30 °C is admirable in the range of 450 kJ

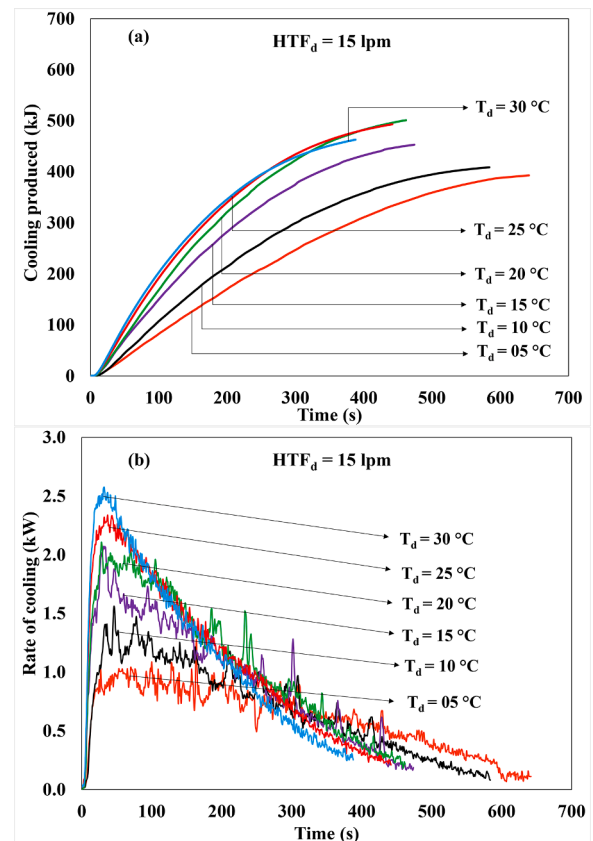


Fig. 7. Variation in (a) cooling produced during desorption and (b) rate of cooling at different desorption temperature. (T_d : Desorption temperature; HTF_d : HTF flow rate during desorption)

Table 2
Cooling produced during desorption

Desorption Temperature (°C)	Amount of hydrogen desorbed (g)	Time taken (s)	Cumulative cooling effect (kJ)	Peak rate of cooling (kW)	Average rate of cooling (kW)
5	45.896	642	393.0	1.3	0.6
10	46.162	584	408.9	1.6	0.7
15	46.894	474	453.0	2.1	1.0
20	48.063	462	501.0	2.1	1.1
25	48.139	442	493.1	2.3	1.1
30	47.729	388	462.9	2.6	1.2

to 500 kJ. The details of the cooling produced is presented in Table 2. The net cooling produced at desorption temperature of 15 °C, 20 °C, 25 °C and 30 °C were 453 kJ, 501 kJ, 493.1 kJ and 462.9 kJ respectively. Nearly 27.5% increase in cumulative cooling effect was observed when the desorption temperature was increased from 5 °C to 20 °C. At desorption temperature of 30 °C, the equilibrium bed pressure was comparatively higher, resulting in rapid desorption during initial 40 s. This rapid desorption further resulted in a higher peak cooling effect, as much as 2.6 kW. The average cooling rate was 1.2 kW while the desorption time was 388 s, which was 16% faster when compared to the duration at desorption temperature of 20 °C.

Though the rate of desorption and cooling were higher, there was slight decrease in the cumulative cooling obtained. With this, highly rapid desorption, the driving force had reduced towards the end of the reaction leading to 0.7% reduction in the amount of H₂ desorbed and 7.4% decrease in cumulative cooling effect. It can be agreed that 462.9 kJ of cooling in 388 s is still an appreciable cooling performance at desorption temperature of 30 °C. Because of rapid cooling under these

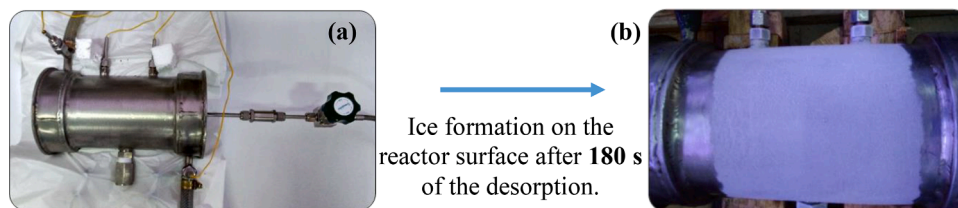


Fig. 8. Pictorial view of reactor (a) before desorption and (b) after 180 s of desorption.

desorption conditions, dense ice formation on the reactor surface was observed, as depicted in Fig. 8 (b). This ice layer was observed after 180 s of the desorption half cycle, which has formed due to the condensation of atmospheric moisture coming in contact with outer surface of reactor during desorption.

Considering the excellent desorption characteristics of $\text{MmNi}_{4.7}\text{Fe}_{0.3}$, this MH reactor can be coupled with the fuel cell stack for delivering hydrogen at required rate with an added advantage of cooling the fuel cell stack. Similarly, it can be used for delivering hydrogen in a fuel cell powered automobile while simultaneously cooling the cabin space.

Conclusions

The effects of H_2 supply pressure and absorption temperature on the absorption kinetics and the effect of desorption temperature on the desorption kinetics of the 55 ECT reactor filled with 4 kg of $\text{MmNi}_{4.7}\text{Fe}_{0.3}$, with water as HTF have been reported in the present study. The experimental observations on the absorption and desorption characteristics from this investigation are as follows:

- $\text{MmNi}_{4.7}\text{Fe}_{0.3}$ possess poor absorption characteristics at ambient temperature and low pressure (less than 50 bar). For $P_s = 70$ bar and $T_a = 25^\circ\text{C}$, the improvement in H_2 storage capacity as compared to P_s of 10, 20, 30, 40, 50 and 60 bar was observed as 11.7, 8.9, 7, 3.2, 1.9 and 1.6 times, respectively.
- Variation in the absorption temperature has a significant impact on the hydriding characteristics of the alloy. At $P_s = 40$ bar, the amount of hydrogen absorbed was comparatively higher for lower T_a . When T_a was decreased from 25°C to 20, 15, 10 and 5°C , the amount of hydrogen absorbed substantially increased by 1.5, 2.6, 3.6 and 3.8 times, respectively.
- Considering this absorption performance, it is recommended to maintain the supply pressure above 60 bar so as to obtain significant absorption of greater than 1 wt.% for T_a in the range of 20°C to 30°C .
- The alloy showed excellent desorption characteristics wherein, the reversible characteristics of the system was nearly 96% at T_d in the ambient range of 20°C to 30°C . At T_d below 20°C , a slight decrease of 4.5% in the amount of desorbed hydrogen has been observed. Nonetheless, even at 5°C , appreciable cooling effect has been obtained i.e 393 kJ in 642 s.
- The cumulative cooling effect produced due to endothermic desorption reaction was evaluated as 501 kJ in desorption half cycle time of 462 s at desorption temperature of 20°C with a HTF flow rate of 15 lpm. At desorption temperature of 30°C , the peak and average rate of cooling produced were as high as 2.6 kW and 1.2 kW, respectively.

Taking note of the excellent desorption characteristics of $\text{MmNi}_{4.7}\text{Fe}_{0.3}$ at near ambient conditions, it can be considered suitable for both stationary and mobile applications. It can be used in fuel cell operated vehicle for continuous supply of hydrogen to the fuel cell and simultaneously providing auxiliary cooling in the automobile cabin. Similarly, it can be integrated for stationary applications, as in for generating energy through fuel cell while cooling the fuel cell stack

simultaneously.

CRediT authorship contribution statement

Alok Kumar: Conceptualization, Methodology, Investigation, Writing - original draft. **Nithin N. Raju:** Investigation, Data curation, Software. **P. Muthukumar:** Validation, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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